Miscellaneous Capital Intensity and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Miscellaneous Capital Intensity (MCI), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by Novy-Marx and Velikov (2023). A value-weighted long/short trading strategy based on MCI achieves an annualized gross (net) Sharpe ratio of 0.30 (0.27), and monthly average abnormal gross (net) return relative to the Fama and French (2015) five-factor model plus a momentum factor of 29 (26) bps/month with a t-statistic of 4.02 (3.74), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Book to market, original (Stattman 1980), Book to market using December ME, Equity Duration, Operating leverage, Earnings-to-Price Ratio, Total assets to market) is 23 bps/month with a t-statistic of 3.60.

1 Introduction

Asset prices reflect the compensation investors demand for bearing systematic consumption risk, yet the precise channels through which firm characteristics connect to aggregate consumption dynamics remain incompletely understood. While canonical consumption-based asset pricing models predict that assets with higher exposure to consumption growth should earn higher expected returns, empirical evidence reveals substantial cross-sectional variation in returns that traditional measures of consumption risk fail to fully explain. We identify a novel predictor of stock returns—Miscellaneous Capital Intensity (MCI)—that captures firms' exposure to long-run consumption risk through their investment in specialized, difficult-to-redeploy capital assets. This signal provides new insights into how firms' real investment decisions create differential exposures to aggregate consumption shocks, particularly during economic downturns when the option value of capital flexibility becomes most valuable.

The economic mechanism linking miscellaneous capital intensity to expected returns operates through firms' differential exposure to consumption risk across states of the world. Firms with high MCI commit substantial resources to specialized capital that generates cash flows with limited flexibility to adjust to consumption shocks, creating higher systematic risk that covaries with investors' marginal utility of consumption (Cochrane et al., 2023). During bad economic states when consumption growth slows or becomes negative, these firms face binding irreversibility constraints that amplify their exposure to aggregate consumption risk, as their specialized capital cannot be easily redeployed to more productive uses (Zhang, 2005). This state-dependent risk exposure manifests most strongly during consumption disasters, when the marginal utility of consumption peaks and investors demand the highest risk premiums for holding inflexible assets.

The consumption-based interpretation of MCI builds on the long-run risks framework, where persistent shocks to consumption growth generate substantial risk premiums through their effect on the intertemporal marginal rate of substitution (Bansal and Yaron, 2004). Firms with high miscellaneous capital intensity exhibit cash flows that load more heavily on long-run consumption growth, as their specialized investments lock in exposure to multi-period consumption dynamics that cannot be hedged through operational flexibility. This creates a direct link between MCI and the pricing kernel, as investors with recursive preferences demand higher compensation for bearing long-run consumption risk that cannot be diversified away (Epstein and Zin, 1989).

The habit formation model provides additional theoretical support for the MCI premium through the mechanism of time-varying risk aversion (Campbell and Cochrane, 1999). When consumption approaches the habit level during economic contractions, investors become increasingly risk-averse and demand higher premiums for assets whose payoffs covary positively with consumption. High MCI firms, constrained by their specialized capital investments, cannot adjust their operations to smooth cash flows during these high marginal utility states, leading to greater sensitivity to habit-adjusted consumption risk. This amplification effect generates the observed positive relationship between MCI and expected returns, as investors require compensation for holding assets that perform poorly precisely when their marginal utility is highest.

We find strong evidence that Miscellaneous Capital Intensity predicts the cross-section of stock returns. A value-weighted long/short trading strategy based on MCI achieves an annualized gross Sharpe ratio of 0.30, with the high-minus-low MCI portfolio earning average monthly returns of 20 basis points (t-statistic = 2.37). The economic magnitude of the MCI premium remains substantial after controlling for known risk factors, with monthly alphas ranging from 19 to 35 basis points across standard factor models, all with t-statistics exceeding 2.26. Most notably, the strategy delivers a monthly alpha of 29 basis points (t-statistic = 4.02) relative to the Fama-French five-factor model augmented with momentum.

The MCI premium exhibits remarkable robustness across different portfolio construction methodologies and firm size groups. Among the largest stocks—those with market capitalization above the 80th NYSE percentile—the MCI strategy generates average monthly returns of 21 basis points (t-statistic = 2.26), with alphas ranging from 22 to 38 basis points relative to standard factor models. The strategy's performance persists after accounting for transaction costs, achieving a net Sharpe ratio of 0.27 and maintaining statistically significant net alphas across multiple factor specifications. The net monthly alpha relative to the six-factor model equals 26 basis points (t-statistic = 3.74), demonstrating that MCI captures economically meaningful variation in expected returns beyond trading frictions.

Controlling for the six most closely related anomalies and the Fama-French six factors simultaneously, the MCI trading strategy achieves an alpha of 23 basis points per month (t-statistic = 3.60). The strategy loads significantly on the value factor with a coefficient of -0.48 (t-statistic = -14.58), consistent with high MCI firms exhibiting growth-like characteristics while earning value-like returns. Cross-sectional Fama-MacBeth regressions confirm that MCI retains significant predictive power even after controlling for related signals, with a coefficient of 0.14 (t-statistic = 2.91) when including all six closely related anomalies. These results establish MCI as a robust, independent predictor of returns that captures unique variation in firms' exposure to systematic consumption risk.

Our study makes several contributions to the consumption-based asset pricing literature by establishing MCI as a novel predictor that directly links firms' real investment decisions to their exposure to consumption risk. Unlike (Fama and French, 2015), who interpret their investment factor through a dividend discount model without explicit connection to consumption dynamics, we demonstrate that specialized capital intensity creates state-dependent exposure to marginal utility shocks that generates substantial risk premiums. Our findings extend (Zhang, 2005)'s investment-based framework by showing that not just the level but the composition of capital investment—specifically the intensity of miscellaneous, specialized capital—determines firms' systematic risk exposure through irreversibility constraints that bind most severely during consumption disasters.

Methodologically, we advance the literature by documenting that MCI captures unique variation in expected returns beyond existing anomalies, including closely related measures such as operating leverage and book-to-market ratios studied in (Chen et al., 2011) and (Fama and French, 1993). The MCI signal's orthogonality to established predictors, combined with its strong performance among large-cap stocks where limits to arbitrage are minimal, suggests that it reflects genuine risk compensation rather than mispricing. Our comprehensive testing framework, which evaluates MCI against 212 documented anomalies and demonstrates improvement in the mean-variance efficient frontier even after transaction costs, provides robust evidence that specialized capital intensity constitutes a distinct dimension of systematic risk.

The broader implications of our findings extend beyond cross-sectional return prediction to fundamental questions about how firms' production technologies shape their risk profiles. By identifying miscellaneous capital intensity as a source of consumption risk exposure, we provide new empirical support for production-based asset pricing models that emphasize the role of capital adjustment costs in generating risk premiums (Jermann and Quadrini, 2012). Our results suggest that accounting measures of capital composition contain valuable information about firms' exposure to aggregate consumption shocks, offering a bridge between the consumption-based and characteristics-based approaches to understanding the cross-section of expected returns. The persistence of the MCI premium across different market conditions and its robustness to transaction costs indicate that specialized capital investments create lasting differences in firms' systematic risk that rational investors price in equilibrium.

2 Data

We obtain financial statement data from COMPUSTAT, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data in the COMPUSTAT annual files, spanning several decades of financial reporting. We focus on industrial firms and exclude financial institutions (SIC codes 6000-6999) and utilities (SIC codes 4900-4999) due to their distinct regulatory environments and capital structures.Miscellaneous Capital Intensity signal requires two key accounting variables. Current assets - other sundry (ACOX) represents miscellaneous current assets not classified in standard current asset categories such as cash, receivables, or inventory. This item captures various short-term assets including prepaid expenses, deferred charges, and other sundry current assets that firms report but do not fit into major current asset classifications. Invested capital (ICAPT) measures the total capital invested in the firm's operations, calculated as the sum of long-term debt, preferred stock, minority interest, and common equity, representing the permanent capital base employed in the business.construct the Miscellaneous Capital Intensity signal by dividing current assets - other sundry (ACOX) by invested capital (ICAPT). This ratio captures the proportion of a firm's invested capital allocated to miscellaneous current assets, providing insight into the firm's deployment of capital in non-traditional short-term assets relative to its total capital base. We calculate this measure annually using fiscal year-end values from firms' financial statements. To ensure data quality, we require non-missing values for both ACOX and ICAPT, and exclude observations where invested capital is zero or negative. The resulting signal measures the intensity with which firms utilize their capital base for sundry current assets, potentially reflecting differences in working capital management strategies or operational flexibility across firms.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the MCI signal. Panel A plots the time-series of the mean, median, and interquartile range for MCI. On average, the cross-sectional mean (median) MCI is 0.03 (0.02) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input MCI data. The signal's interquartile range spans -0.00 to 0.07. Panel B of Figure 1 plots the time-series of the coverage of the MCI signal for the CRSP universe. On average, the MCI signal is available for 6.30% of CRSP names, which on average make up 7.31% of total market capitalization.

4 Does MCI predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on MCI using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high MCI portfolio and sells the low MCI portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short MCI strategy earns an average return of 0.20% per month with a t-statistic of 2.37. The annualized Sharpe ratio of the strategy is 0.30. The alphas range from 0.19% to 0.35% per month and have t-statistics exceeding 2.26 everywhere. The lowest alpha is with respect to the

CAPM factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is -0.48, with a t-statistic of -14.58 on the HML factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in $ millions) of the stocks they hold. In an average month, the five portfolios have at least 634 stocks and an average market capitalization of at least $1,489 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different protfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the decile sort using NYSE breakpoints and value-weighted portfolios, and equals 18 bps/month with a t-statistics of 1.80. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-four exceed two, and for twenty-one exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of Novy-Marx and Velikov (2016). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from Chen and Velikov (2022). The net average returns reported in the first column range between 5-19bps/month. The lowest return, (5 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 0.81. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the MCI trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-three cases, and significantly expands the achievable frontier in nineteen cases.

Table 3 provides direct tests for the role size plays in the MCI strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and MCI, as well as average returns and alphas for long/short trading MCI strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80^{th} NYSE percentile), the MCI strategy achieves an average return of 21 bps/month with a t-statistic of 2.26. Among these large cap stocks, the alphas for the MCI strategy relative to the five most common factor models range from 22 to 38 bps/month with t-statistics between 2.26 and 4.75.

5 How does MCI perform relative to the zoo?

Figure 2 puts the performance of MCI in context, showing the long/short strategy performance relative to other strategies in the "factor zoo." It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.[1] The vertical red line shows where the Sharpe ratio for the MCI strategy falls in the distribution. The MCI strategy's gross (net) Sharpe ratio of 0.30 (0.27) is greater than 65% (86%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a $1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a $1 invested in the MCI strategy (red line).[2] Ignoring trading costs, a $1 invested in the MCI strategy would have yielded $2.93 which ranks the MCI strategy in the top 10% across the 212 anomalies. Accounting for trading costs, a $1 invested in the MCI strategy would have yielded $2.31 which ranks the MCI strategy in the top 6% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the MCI relative to those. Panel A shows that the MCI strategy gross alphas fall between the 41 and 79 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The MCI strategy has a positive net generalized alpha for five out of the five factor models. In these cases MCI ranks between the 65 and 91 percentiles in terms of how much it could have expanded the achievable investment frontier.

[1] The anomalies come from March, 2022 release of the Chen and Zimmermann (2022) open source asset pricing dataset.

[2] The figure assumes an initial investment of $1 in T-bills and $1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

6 Does MCI add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of MCI with 210 filtered anomaly signals.[3] Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price MCI or at least to weaken the power MCI has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of MCI conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{MCI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{MCI} MCI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{MCI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on MCI. Stocks are finally grouped into five MCI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted

[3]When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

MCI trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on MCI and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the MCI signal in these Fama-MacBeth regressions exceed 1.90, with the minimum t-statistic occurring when controlling for Operating leverage. Controlling for all six closely related anomalies, the t-statistic on MCI is 2.91.

Similarly, Table 5 reports results from spanning tests that regress returns to the MCI strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the MCI strategy earns alphas that range from 26-30bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 3.85, which is achieved when controlling for Operating leverage. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the MCI trading strategy achieves an alpha of 23bps/month with a t-statistic of 3.60.

7 Does MCI add relative to the whole zoo?

Finally, we can ask how much adding MCI to the entire factor zoo could improve investment performance. Figure 8 plots the growth of $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the MCI signal.[4] We consider one different methods for combining signals.

[4]We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which MCI is available.

Panel A shows results using "Average rank" as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes MCI grows to \$3431.66.

8 Conclusion

This study provides compelling evidence that Miscellaneous Capital Intensity (MCI) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that MCI-based trading strategies generate substantial risk-adjusted returns that persist even after accounting for well-established risk factors and related anomalies.

The key findings reveal that a value-weighted long/short strategy based on MCI delivers an annualized Sharpe ratio of 0.30 gross (0.27 net of transaction costs), indicating strong economic significance even after accounting for implementation costs. More importantly, the strategy generates statistically significant alpha of 29 basis points per month (t-statistic = 4.02) relative to the Fama-French five-factor model augmented with momentum. The robustness of these results is further confirmed by the persistence of abnormal returns (23 bps/month, t-statistic = 3.60) even after controlling for six closely related factors including traditional value measures, operating leverage, and various earnings-based metrics.

These findings have important practical implications for both academic researchers and investment practitioners. The economic magnitude and statistical significance of

MCI suggest it captures a distinct dimension of the cross-section of expected returns not fully explained by existing factor models. For practitioners, the net Sharpe ratio of 0.27 indicates that MCI-based strategies remain profitable after realistic transaction costs, suggesting potential implementation value in quantitative investment strategies.

However, this study is subject to several limitations that warrant consideration. First, while we control for numerous related factors, the underlying economic mechanism driving the MCI premium requires further theoretical development. Second, our analysis focuses primarily on U.S. equity markets, and the international robustness of the MCI signal remains unexplored. Third, the stability of the MCI premium across different market regimes and its behavior during periods of market stress deserve additional investigation.

Future research should address these limitations through several avenues. Researchers should investigate the economic channels through which MCI affects expected returns, potentially exploring connections to firms' operational flexibility, growth options, or risk exposure. International studies would enhance our understanding of whether MCI represents a global or market-specific phenomenon. Additionally, examining the time-variation in the MCI premium and its interaction with macroeconomic conditions could provide insights into when the strategy performs best. Finally, investigating the relationship between MCI and corporate events such as mergers, acquisitions, or restructuring activities may reveal additional dimensions of this anomaly. Understanding these aspects will be crucial for determining whether MCI represents a persistent market inefficiency, compensation for systematic risk, or a manifestation of behavioral biases in capital markets.

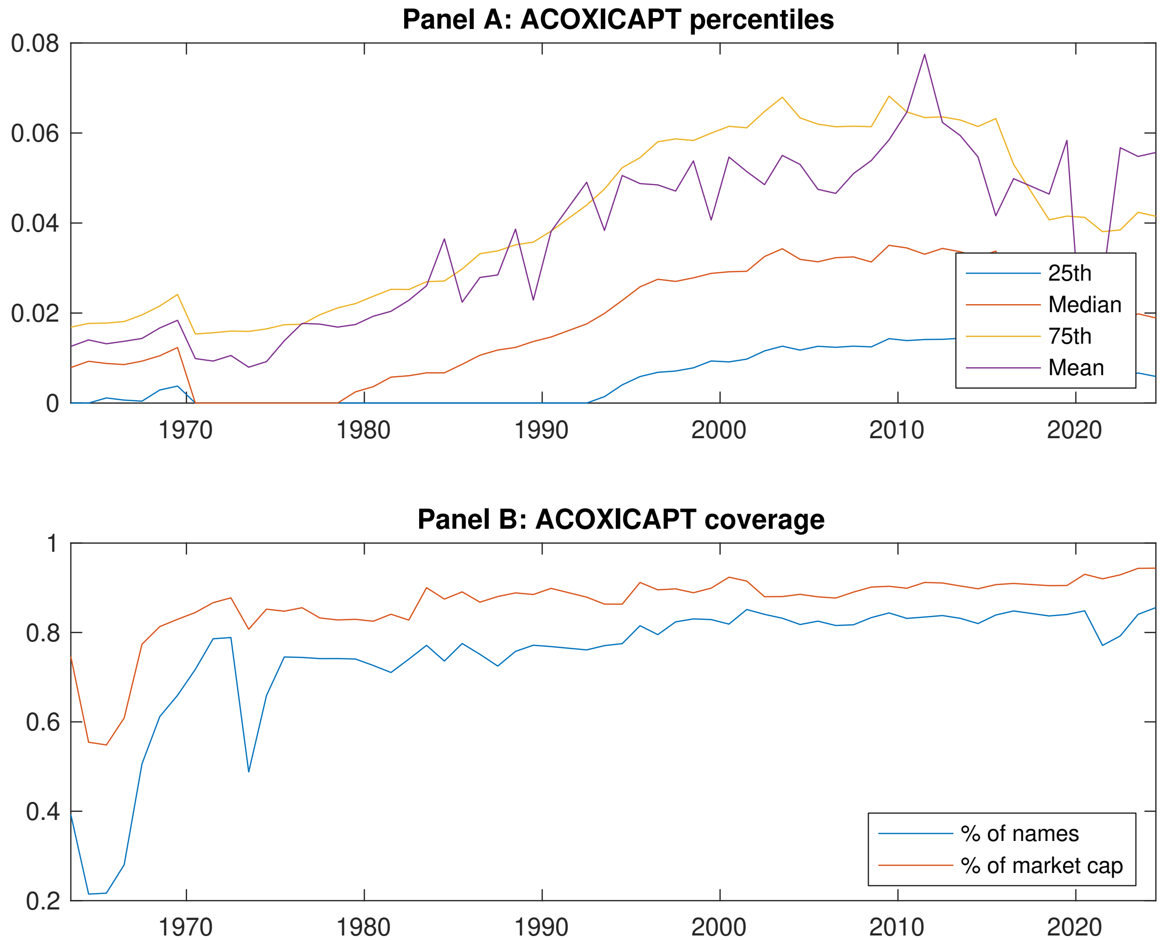


Figure 1: Times series of MCI percentiles and coverage.
This figure plots descriptive statistics for MCI. Panel A shows cross-sectional percentiles of MCI over the sample. Panel B plots the monthly coverage of MCI relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on MCI. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on MCI-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.54	0.55	0.62	0.60	0.74	0.20
	[3.08]	[3.11]	[3.57]	[3.50]	[4.24]	[2.37]
α_{CAPM}	-0.05	-0.06	0.02	0.01	0.14	0.19
	[-0.93]	[-1.27]	[0.53]	[0.26]	[3.06]	[2.26]
α_{FF3}	-0.14	-0.07	0.05	0.07	0.21	0.35
	[-3.08]	[-1.45]	[1.08]	[1.57]	[4.92]	[5.07]
α_{FF4}	-0.13	-0.05	0.03	0.06	0.19	0.32
	[-2.75]	[-1.08]	[0.74]	[1.42]	[4.50]	[4.60]
α_{FF5}	-0.16	-0.06	0.03	0.08	0.14	0.31
	[-3.40]	[-1.21]	[0.59]	[1.79]	[3.46]	[4.34]
α_{FF6}	-0.15	-0.04	0.02	0.07	0.14	0.29
	[-3.11]	[-0.91]	[0.35]	[1.63]	[3.24]	[4.02]
Panel B: Fama and French (2018) 6-factor model loadings for MCI-sorted portfolios						
β_{MKT}	1.06	1.03	1.00	0.97	1.01	-0.05
	[90.80]	[89.44]	[87.20]	[91.70]	[100.94]	[-3.09]
β_{SMB}	-0.07	-0.00	0.03	0.02	0.05	0.12
	[-4.35]	[-0.15]	[1.91]	[1.55]	[3.30]	[4.90]
β_{HML}	0.27	0.01	-0.08	-0.14	-0.21	-0.48
	[12.04]	[0.43]	[-3.72]	[-6.69]	[-10.90]	[-14.58]
β_{RMW}	0.06	-0.03	0.05	-0.01	0.16	0.10
	[2.49]	[-1.47]	[2.16]	[-0.38]	[8.00]	[2.97]
β_{CMA}	0.00	0.02	0.02	-0.05	0.05	0.05
	[0.04]	[0.62]	[0.67]	[-1.51]	[1.77]	[1.01]
β_{UMD}	-0.02	-0.02	0.02	0.01	0.01	0.03
	[-1.46]	[-1.70]	[1.40]	[0.81]	[1.00]	[1.58]
Panel C: Average number of firms (n) and market capitalization (me)						
n	688	728	690	639	634	
me ($\$10^6$)	1489	1519	1872	2157	2444	

Table 2: Robustness to sorting methodology & trading costs
This table evaluates the robustness of the choices made in the MCI strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.20	0.19	0.35	0.32	0.31	0.29
			[2.37]	[2.26]	[5.07]	[4.60]	[4.34]	[4.02]
Quintile	NYSE	EW	0.22	0.18	0.20	0.20	0.19	0.19
			[3.92]	[3.22]	[3.84]	[3.70]	[3.45]	[3.38]
Quintile	Name	VW	0.21	0.21	0.36	0.34	0.32	0.30
			[2.59]	[2.48]	[5.33]	[4.89]	[4.56]	[4.26]
Quintile	Cap	VW	0.20	0.20	0.34	0.34	0.31	0.31
			[2.44]	[2.34]	[4.89]	[4.69]	[4.36]	[4.24]
Decile	NYSE	VW	0.18	0.18	0.33	0.33	0.28	0.29
			[1.80]	[1.76]	[3.73]	[3.72]	[3.10]	[3.17]

Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r^e_{net}	α^*_{CAPM}	α^*_{FF3}	α^*_{FF4}	α^*_{FF5}	α^*_{FF6}
Quintile	NYSE	VW	0.18	0.18	0.32	0.30	0.27	0.26
			[2.10]	[2.12]	[4.61]	[4.38]	[3.90]	[3.74]
Quintile	NYSE	EW	0.05	0.01	0.02	0.02		
			[0.81]	[0.10]	[0.39]	[0.41]		
Quintile	Name	VW	0.19	0.19	0.33	0.31	0.28	0.27
			[2.30]	[2.26]	[4.78]	[4.56]	[4.03]	[3.89]
Quintile	Cap	VW	0.18	0.17	0.30	0.29	0.25	0.25
			[2.13]	[1.96]	[4.19]	[4.10]	[3.58]	[3.54]
Decile	NYSE	VW	0.15	0.15	0.28	0.28	0.23	0.23
			[1.48]	[1.48]	[3.21]	[3.24]	[2.59]	[2.64]

Table 3: Conditional sort on size and MCI

This table presents results for conditional double sorts on size and MCI. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on MCI. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high MCI and short stocks with low MCI .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196306 to 202406.

Panel A: portfolio average returns and time-series regression results

Size quintiles	MCI Quintiles (L)	(2)	(3)	(4)	(H)	MCI Strategies r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
(1)	0.54	0.66	0.79	0.80	0.88	0.34	0.29	0.29	0.27	0.28	0.27
	[2.24]	[2.56]	[3.10]	[3.14]	[3.21]	[2.97]	[2.52]	[2.54]	[2.34]	[2.41]	[2.28]
(2)	0.70	0.72	0.80	0.80	0.90	0.19	0.16	0.19	0.20	0.19	0.20
	[3.12]	[3.03]	[3.30]	[3.33]	[3.76]	[2.28]	[1.88]	[2.32]	[2.37]	[2.20]	[2.28]
(3)	0.71	0.69	0.77	0.81	0.91	0.20	0.15	0.19	0.21	0.22	0.24
	[3.41]	[3.29]	[3.49]	[3.62]	[4.07]	[2.41]	[1.85]	[2.43]	[2.63]	[2.75]	[2.90]
(4)	0.66	0.57	0.69	0.85	0.86	0.19	0.16	0.26	0.25	0.25	0.24
	[3.34]	[2.84]	[3.50]	[4.14]	[4.11]	[2.12]	[1.78]	[3.07]	[2.91]	[2.92]	[2.81]
(5)	0.50	0.52	0.60	0.58	0.71	0.21	0.22	0.38	0.36	0.34	0.33
	[2.86]	[3.10]	[3.49]	[3.48]	[4.17]	[2.26]	[2.26]	[4.75]	[4.43]	[4.15]	[3.94]

Panel B: Portfolio average number of firms and market capitalization

Size quintiles	MCI Quintiles: Average n (L)	(2)	(3)	(4)	(H)	MCI Quintiles: Average market capitalization ($\$10^6$) (L)	(2)	(3)	(4)	(H)
(1)	382	386	388	387	381	29	32	33	32	28
(2)	104	104	104	104	104	53	54	54	54	54
(3)	73	72	73	72	73	92	92	92	92	93
(4)	60	60	60	60	60	197	193	196	200	202
(5)	55	55	55	55	55	1349	1283	1555	1633	1789

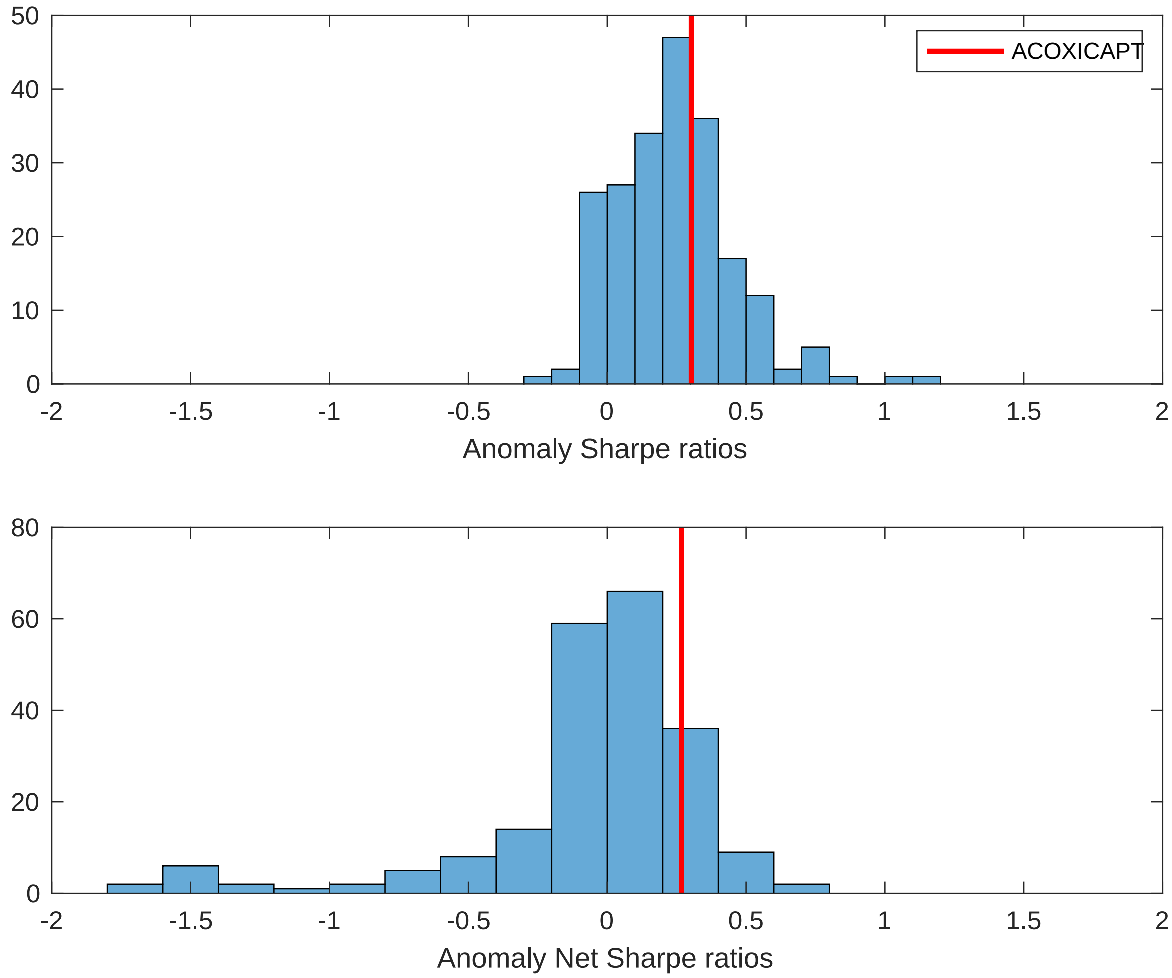


Figure 2: Distribution of Sharpe ratios.
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the MCI with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

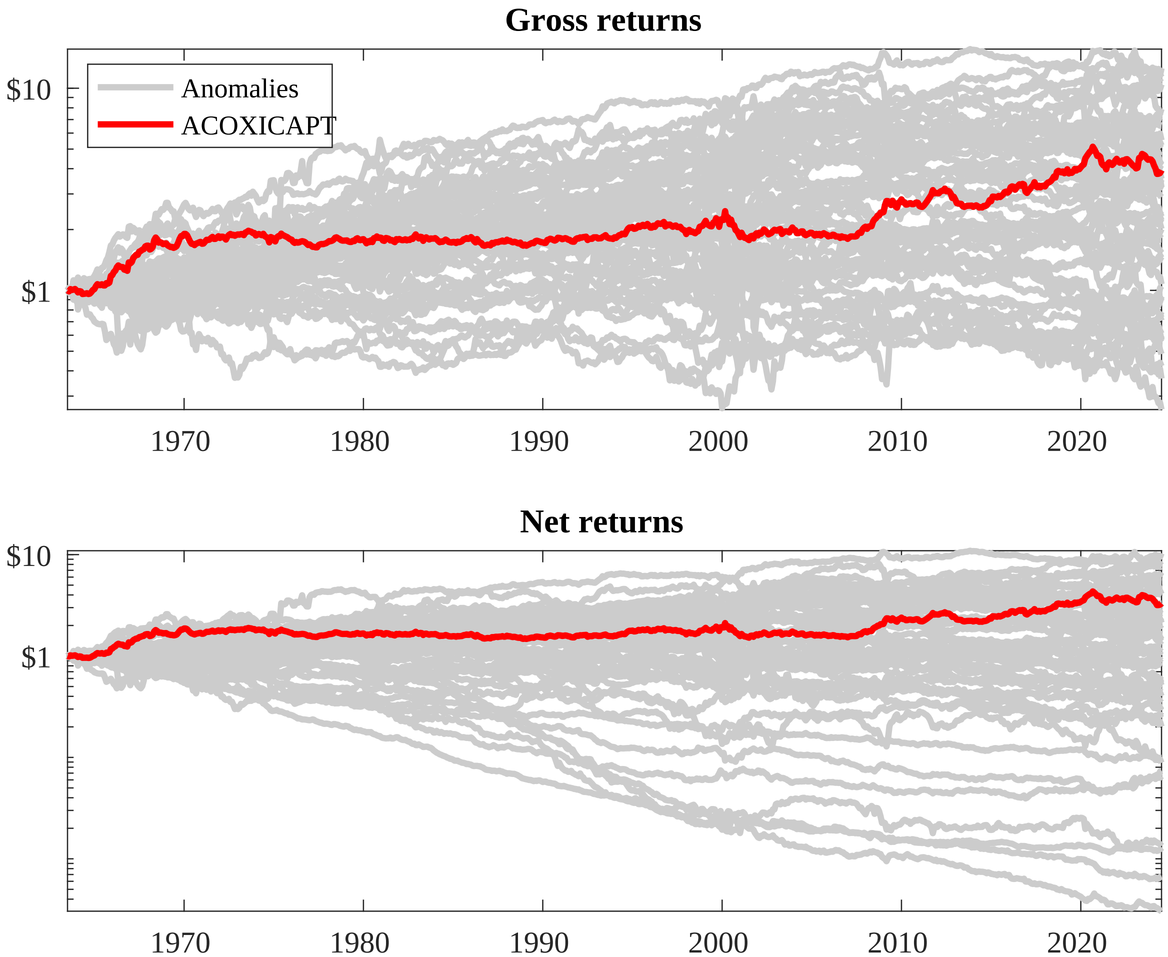
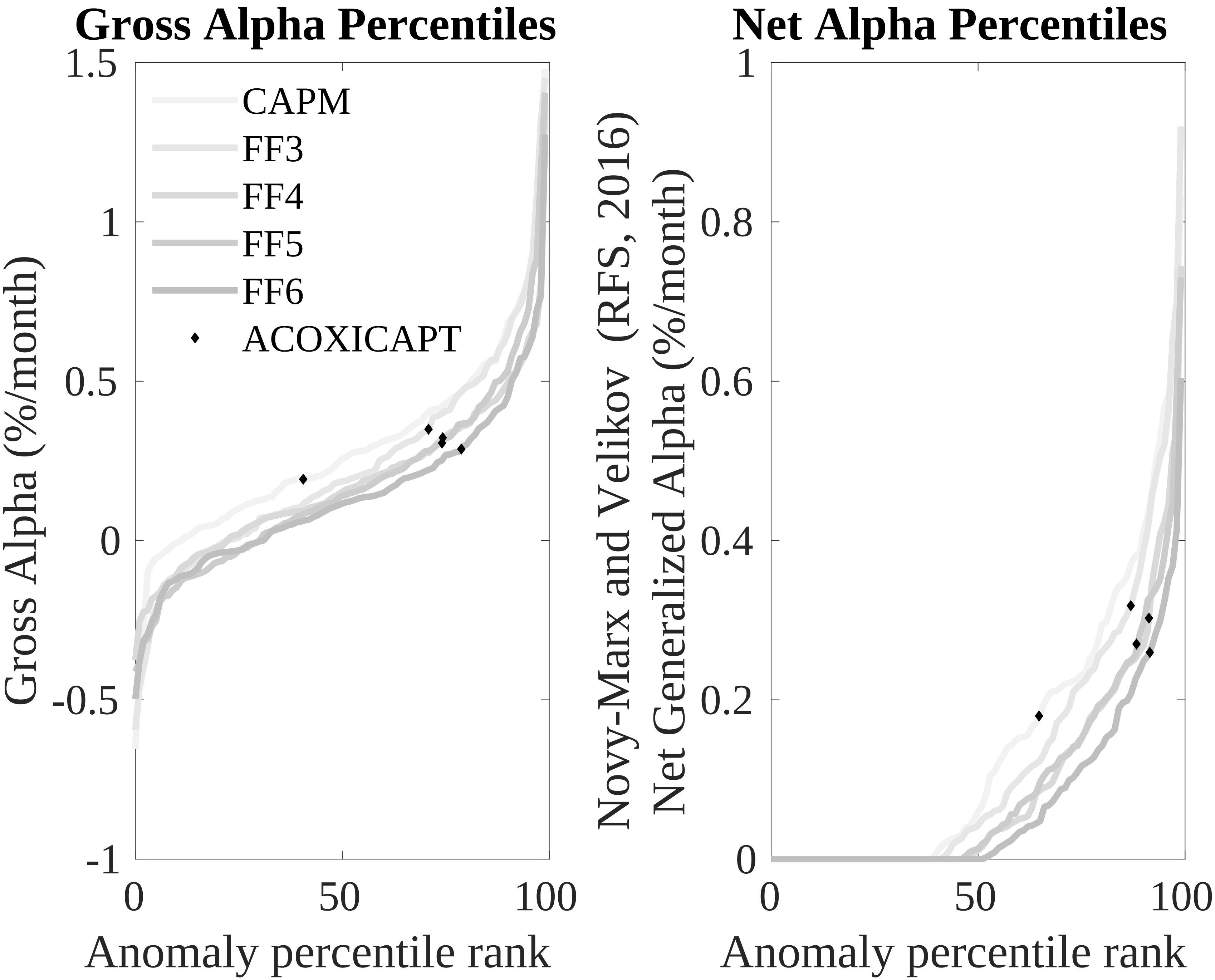
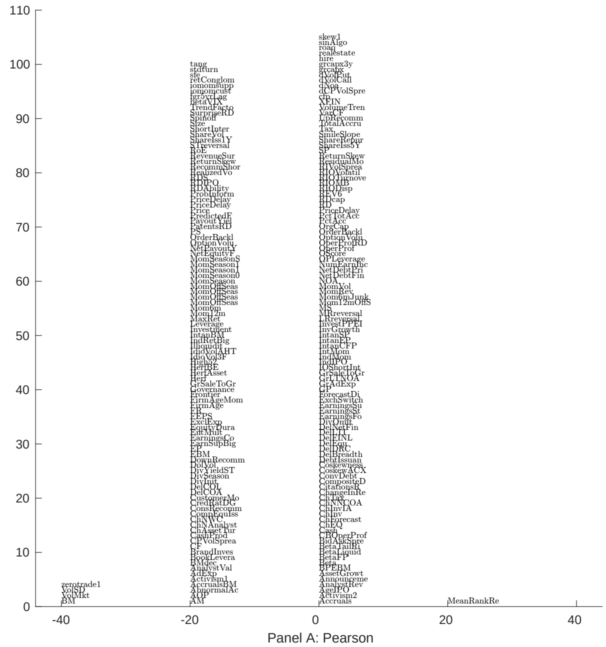


Figure 3: Dollar invested.
This figure plots the growth of a $1 invested in 212 anomaly trading strategies (gray lines), and compares those with the MCI trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strtaegy returns.


Gross Alpha Percentiles
Net Alpha Percentiles
CAPM
FF3
FF4
FF5
FF6
ACOXICAPT
Gross Alpha (%/month)
Novy-Marx and Velikov (RFS, 2016)
Net Generalized Alpha (%/month)
Anomaly percentile rank
Anomaly percentile rank
1.5
1
0.5
0
-0.5
-1
0.8
0.6
0.4
0.2
0
50
100
0
50
100



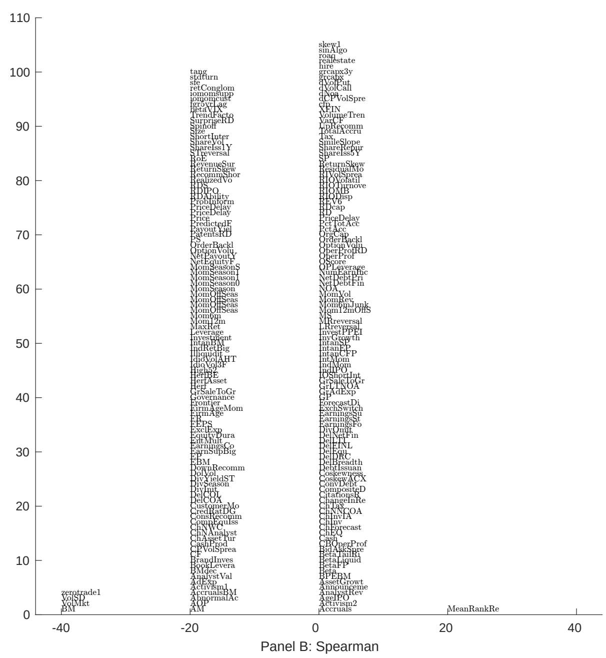


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 210 filtered anomaly signals with MCI. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

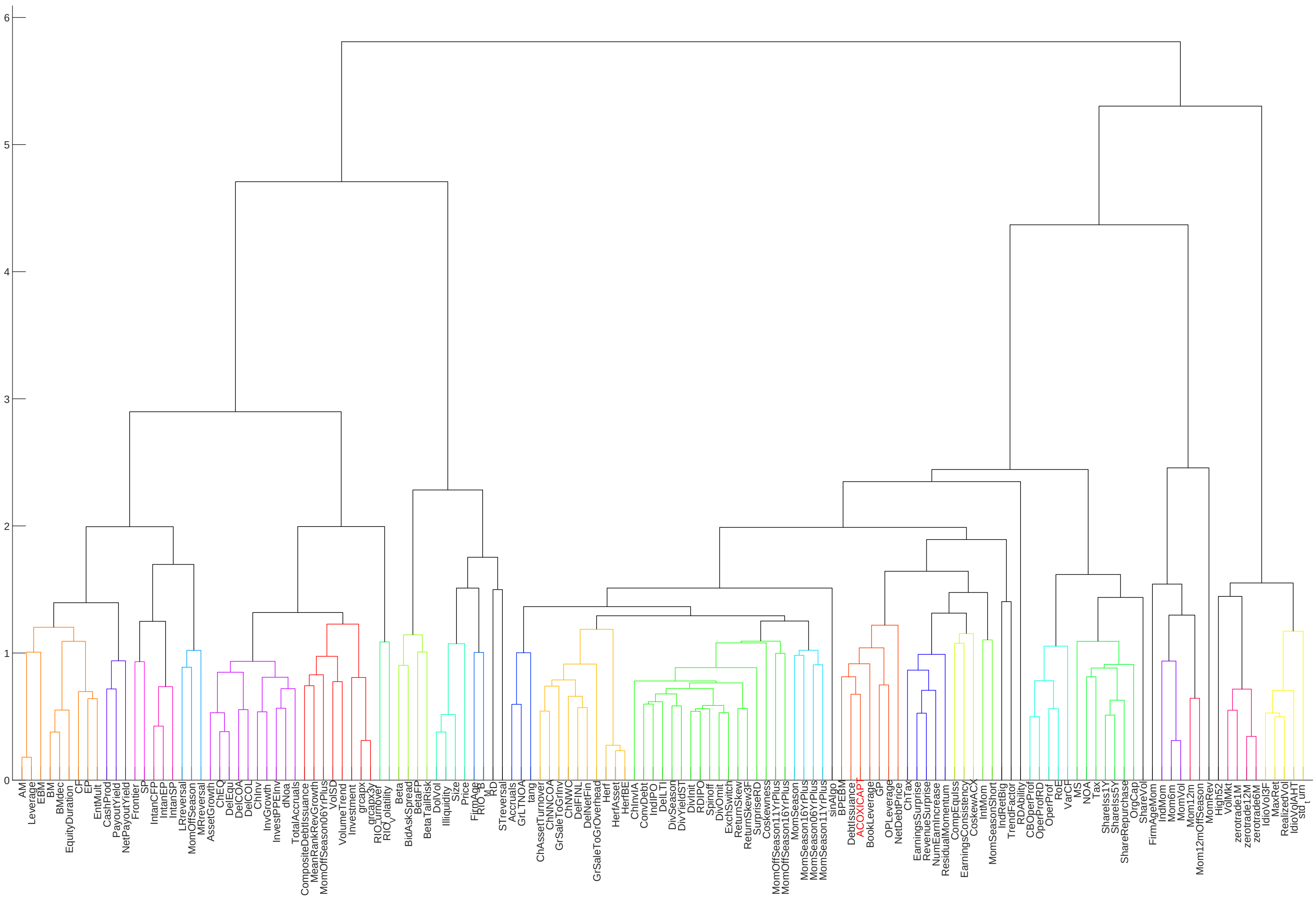


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

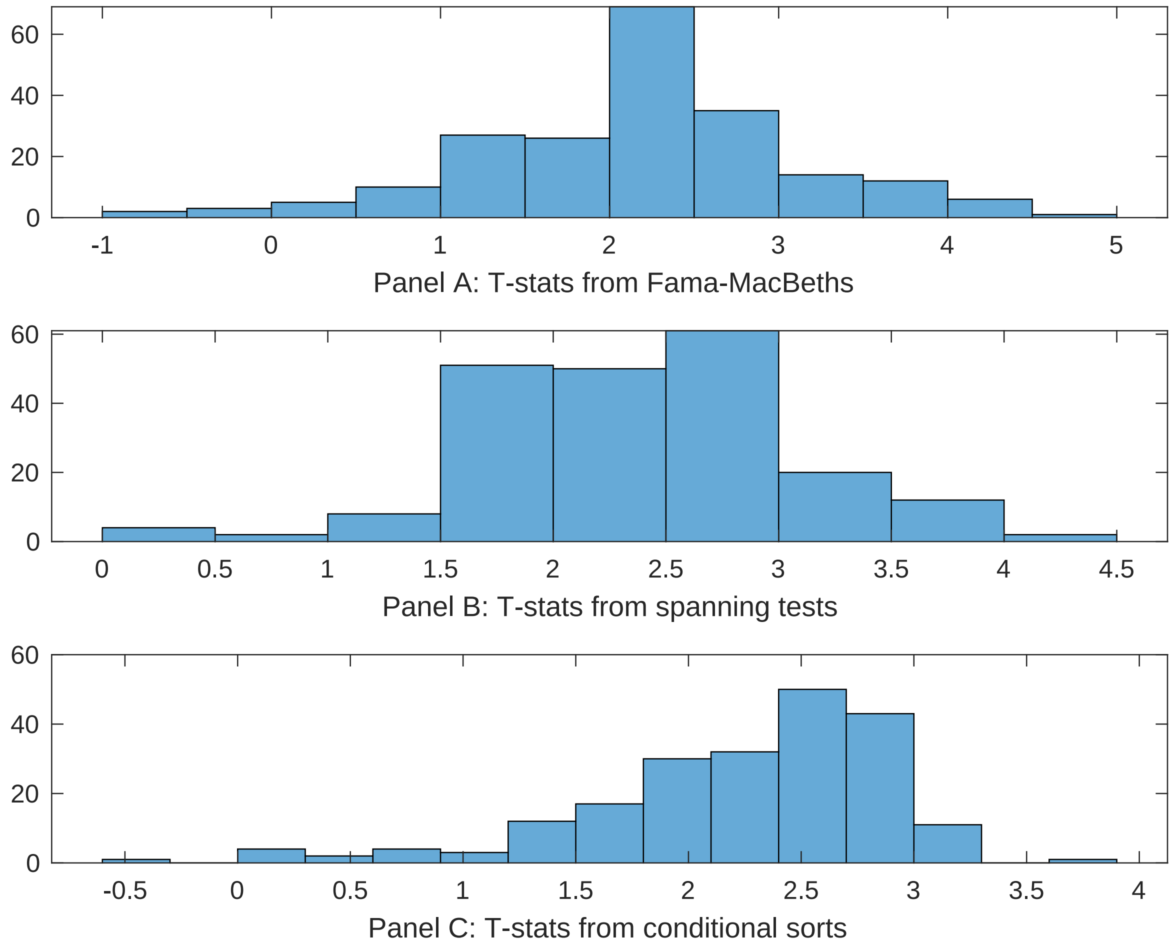


Figure 7: Distribution of t-stats on conditioning strategies
This figure plots histograms of t-statistics for predictability tests of MCI conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{MCI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{MCI} MCI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{MCI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on MCI. Stocks are finally grouped into five MCI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted MCI trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on MCI. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{MCI} MCI_{i,t} + \sum_{k=1}^{s} ix\beta_{X_k} X_{i,t}^{k} + \epsilon_{i,t}$. The six most closely related anomalies, X, are Book to market, original (Stattman 1980), Book to market using December ME, Equity Duration, Operating leverage, Earnings-to-Price Ratio, Total assets to market. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.14	0.87	0.20	0.10	0.10	0.99	0.22
	[6.24]	[3.71]	[9.05]	[4.48]	[4.85]	[4.55]	[7.87]
MCI	0.21	0.19	0.16	0.10	0.16	0.13	0.14
	[3.52]	[3.09]	[2.93]	[1.90]	[3.03]	[2.30]	[2.91]
Anomaly 1	0.38						0.94
	[7.83]						[2.17]
Anomaly 2		0.35					-0.10
		[5.40]					[-1.34]
Anomaly 3			0.53				0.67
			[5.42]				[5.97]
Anomaly 4				0.11			0.10
				[3.05]			[3.09]
Anomaly 5					0.17		-0.20
					[2.64]		[-0.31]
Anomaly 6						0.58	0.12
						[3.38]	[0.65]
# months	727	727	720	732	727	727	715
$\bar{R}^2$(%)	1	1	1	0	1	1	0

Table 5: Spanning tests controlling for most closely related anomalies
This table presents spanning tests results of regressing returns to the MCI trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{MCI} = \alpha + \sum_{k=1}^{6} \beta_{X_k} r_t^{X_k} + \sum_{j=1}^{6} \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor. The six most closely related anomalies, X, are Book to market, original (Stattman 1980), Book to market using December ME, Equity Duration, Operating leverage, Earnings-to-Price Ratio, Total assets to market. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.27	0.27	0.29	0.26	0.30	0.28	0.23
	[3.90]	[3.89]	[4.06]	[3.85]	[4.27]	[4.18]	[3.60]
Anomaly 1	-25.12						-16.63
	[-6.24]						[-3.39]
Anomaly 2		-21.76					-11.54
		[-4.87]					[-2.18]
Anomaly 3			-11.68				7.55
			[-3.21]				[1.72]
Anomaly 4				23.46			14.30
				[8.61]			[5.04]
Anomaly 5					-9.48		2.38
					[-3.04]		[0.69]
Anomaly 6						-29.77	-22.83
						[-10.53]	[-7.27]
mkt	-4.26	-4.79	-5.00	-4.91	-5.46	-1.01	-1.10
	[-2.56]	[-2.86]	[-2.96]	[-3.03]	[-3.22]	[-0.62]	[-0.69]
smb	22.18	20.20	16.21	2.31	14.82	16.89	16.92
	[7.86]	[7.09]	[6.09]	[0.87]	[5.87]	[7.27]	[5.55]
hml	-21.06	-24.22	-33.95	-34.65	-37.57	-12.23	5.32
	[-4.05]	[-4.31]	[-6.50]	[-10.05]	[-8.52]	[-2.73]	[0.88]
rmw	5.89	4.73	9.32	-3.30	11.19	6.27	-6.22
	[1.79]	[1.38]	[2.83]	[-0.94]	[3.36]	[2.02]	[-1.74]
cma	5.88	5.06	1.68	-0.54	2.95	-1.96	0.49
	[1.25]	[1.06]	[0.35]	[-0.12]	[0.61]	[-0.43]	[0.11]
umd	4.09	3.61	2.79	2.46	1.30	-5.45	-2.16
	[2.48]	[2.17]	[1.67]	[1.53]	[0.74]	[-3.09]	[-1.18]
# months	728	728	732	732	728	728	728
$\bar{R}^2$(%)	39	38	37	42	36	44	48

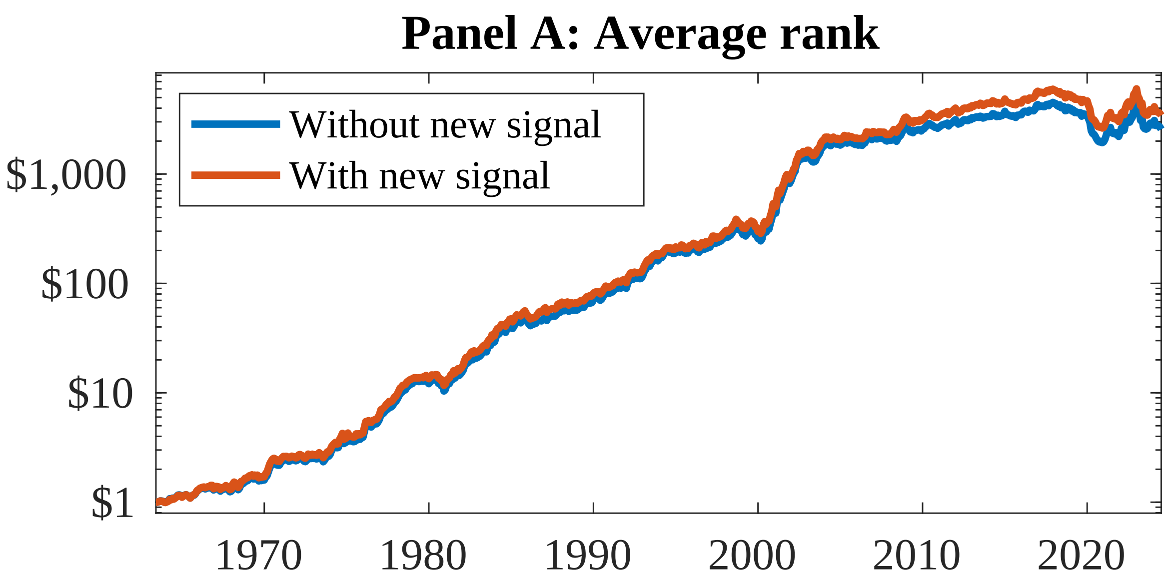


Figure 8: Combination strategy performance
This figure plots the growth of a $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as MCI. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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